Daily Logic Drill #7

Sheet by The-CerealDev

Section	**Topic**	**Questions**	**Target time**
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mixed Synthesis Capstone · Every technique from Days 1–6, combined.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

One quick question per technique from the whole week. Pure recognition, full speed.

1. State the contrapositive of: "If a sequence is convergent, then it is bounded."
2. What is the negation of: "No integer is both even and odd"?
3. Is "n is a multiple of 6" necessary, sufficient, both, or neither for "n is a multiple of 2"?
4. To prove "there is no largest prime number" by contradiction, what should you assume first?
5. Give a counterexample to: "Every odd number is prime."
6. Name the fallacy: "If it is raining, the ground is wet. The ground is wet. Therefore it is raining."
7. What is the converse of "If p then q"?
8. To prove directly that "if n is a multiple of 7, then n^2 is a multiple of 49", what is the very first line of the proof?
9. True or false: "$\exists x\, \forall y\, (x < y)$ and $\forall y\, \exists x\, (x < y)$ are logically equivalent statements about real numbers."
10. Which proof technique is most natural for disproving "every multiple of 3 is odd"?

Section B Manipulation Drills (≤50 s each)

Each question pairs two techniques from different days.

1. Prove by contrapositive: "If n is not a multiple of 3 and not a multiple of 5, then n is not a multiple of 15."
2. Prove by contradiction: "There is no positive integer n such that $n^2 + n$ is odd."
3. Prove by induction: $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all positive integers n.

4. A student wants to prove "if n is not a multiple of 5, then n^2 is not a multiple of 25" by instead showing "if n is a multiple of 5, then n^2 is a multiple of 25." Is this a valid proof of the original claim? Justify your answer.

5. Give a counterexample showing that "n is prime" is not sufficient for "n is odd."

6. Negate the statement: "For every set A, there exists a set B such that $A \cap B = \emptyset$." Is the negated statement true or false?

7. Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers n.

8. Prove by contradiction that $\sqrt{6}$ is irrational.

9. For a polynomial f, classify "$f(x) > 0$ for all real x" as necessary, sufficient, both, or neither for "f has no real roots."

10. A student proves $P(n)$ for all $n \geq 1$ by checking $P(1)$ directly, then showing $P(k) \implies P(k+1)$ only for $k \geq 2$ (skipping $k = 1$). Explain the gap this leaves in the proof.

Section C Substitution & Structure (≤90 s each)

Tricky synthesis MCQs, combining techniques from across the week.

1. Let P: "n is divisible by 4" and Q: "n is divisible by 2 and n is divisible by 6". Which of the following is true?

A) $P \iff Q$.

B) P is sufficient but not necessary for Q.

C) P is necessary but not sufficient for Q.

D) P and Q are neither necessary nor sufficient for each other.

E) Q is impossible to satisfy.

2. A theorem states: "If every element of a finite set S of positive integers is even, then the sum of all elements of S is even." A student, wanting to prove the converse, argues: "Take $S = \{2, 4\}$; the sum is 6, which is even, and every element is indeed even, confirming the converse." What is the flaw?

A) The theorem itself is false.

B) A single supporting example cannot establish a universal converse statement.

C) $S = \{2, 4\}$ does not actually have an even sum.

D) The original theorem's proof technique (not the converse) is what's flawed.

E) There is no flaw; the converse is proved.

3. Which proof technique is most directly suited to establishing: "There exists a positive integer n such that $n^2 + 3n + 1$ is not prime"?

A) Induction.

B) Direct exhibition of a specific such n (e.g. $n = 6$: $6^2 + 3(6) + 1 = 55 = 5 \times 11$).

C) Contradiction.

D) Contrapositive.

E) Strong induction.

4. A student claims: "n is a multiple of 9" is necessary and sufficient for "the digit sum of n is a multiple of 9". They verify this only for $n = 9, 18, 27$ and declare the claim proved. Evaluate this.

A) The claim and the verification are both fully correct.

B) Checking three cases does not constitute a proof of a necessary-and-sufficient claim holding for every positive integer n.

C) The claim is false; digit sums have nothing to do with divisibility by 9.

D) The claim is true only for necessity, not sufficiency.

E) The claim is true only for sufficiency, not necessity.

5. What is the contrapositive of "If n^2 is not divisible by 4, then n is odd"?

A) If n is odd, then n^2 is not divisible by 4.

B) If n is even, then n^2 is divisible by 4.

C) If n^2 is divisible by 4, then n is even.

D) If n is not odd, then n^2 is not divisible by 4.

E) If n^2 is divisible by 4, then n is odd.

6. A student claims $P(n)$: "$n^2 + 5n + 2$ is even for every positive integer n" is provable by induction: base case $P(1)$: $1 + 5 + 2 = 8$ is even; inductive step: assume $k^2 + 5k + 2$ is even; then $(k+1)^2 + 5(k+1) + 2 = k^2 + 7k + 8 = (k^2 + 5k + 2) + 2k + 6$, which is (even)+(even)=even, so $P(k+1)$ holds. Is this induction valid, and is the underlying claim true?

A) The induction is valid, and the claim is true for every positive integer.

B) The induction has a flaw in the algebra.

C) The claim is false — there exists some n for which $n^2 + 5n + 2$ is odd.

D) The base case is wrong.

7. Consider positive integers m, n. Statement P: "For every prime p, if $p \mid mn$ then $p \mid m$ or $p \mid n$." Statement Q: "There exists a prime p such that $p \mid mn$ but $p \nmid m$ and $p \nmid n$." Which of the following is true?

A) P and Q can both be true simultaneously.

B) Q is the negation of P, so Q is always false because P is always true.

C) P is the negation of Q.

D) Neither P nor Q is ever true.

E) P is false in general.

8. A student claims: "$x = 1$" is necessary and sufficient for "$x^3 - 1 = 0$" (for real x), reasoning: "Sufficient, because substituting $x = 1$ gives 0. Necessary, because $1^3 - 1 = 0$, which confirms $x = 1$ works." Evaluate the necessity argument specifically.

A) Both parts of the reasoning are fully correct as given.

B) The "necessity" argument is circular: showing $1^3 - 1 = 0$ just re-verifies sufficiency again, and says nothing about whether $x = 1$ is the ONLY solution.

C) The necessity argument is flawed because $x = -1$ is also a solution.

D) Sufficiency is actually false.

E) There is no correct way to establish necessity here.

Section D Challenge (SMC / BMO1 difficulty)

Full capstone proofs, drawing on every technique from the week.

1. Long John Silverman has buried treasure at an integer point (x, y) (not necessarily positive). He records the two values $x^2 + y$ and $x + y^2$; these two recorded numbers are distinct from each other. Prove that no other integer point $(x', y') \neq (x, y)$ can produce the same pair of recorded values (i.e. satisfy both $x'^2 + y' = x^2 + y$ and $x' + y'^2 = x + y^2$) — so the map pins down the treasure's location uniquely. *(after BMO1 2010 Q6)*

2. Two bags contain m and n balls respectively $(m, n > 0)$. Two operations are allowed, applied any number of times in any order: (a) remove the same number of balls from each bag; (b) triple the number of balls in one (chosen) bag. *(after BMO1 2012 Q4)*
Prove that if $m - n$ is odd, it is impossible to ever empty both bags (i.e. reach $(0, 0)$) using only these two operations.

3. Prove that there do not exist positive integers $a < b < c$ with $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$ and a, b, c all even.

4. Prove, using strong induction, that every positive integer $n \geq 2$ can be written as a product of primes in exactly one way, up to the order of the factors (the uniqueness part of the Fundamental Theorem of Arithmetic) — assuming as given Euclid's Lemma: if a prime p divides a product ab, then $p \mid a$ or $p \mid b$.

5. Consider the statement $(*)$: "For all positive integers m, n: m and n are coprime if and only if there exist integers x, y (not necessarily positive) with $mx + ny = 1$" (Bézout's Identity).
(a) Prove the $\Longleftarrow$ direction of $(*)$ directly.
(b) Taking the $\Longrightarrow$ direction of $(*)$ as given (it is normally proved via the Euclidean algorithm), use $(*)$ to prove: if p is prime and $p \nmid a$, then there exist integers x, y with $px + ay = 1$; hence deduce Euclid's Lemma (if $p \mid ab$ then $p \mid a$ or $p \mid b$) as a consequence.

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths